

Abhishek Singh

CS undergrad at KIIT building end-to-end ML systems — from EDA and model training to deployment. Experienced with Python, Scikit-learn, TensorFlow, and FastAPI.

Seeking Data Science / Data Analytics internships
[gmail](#) | [portfolio](#) | [linkedin](#) | [github](#) | [itc.io](#)

EDUCATION

Kalinga Institute of Industrial Technology (KIIT)

Bachelor of Technology in Computer Science and Engineering

Bhubaneswar, India

2023 – Present | CGPA: 8.48

PROJECTS

- Talk with CSV** | *React, FastAPI, Groq AI (Llama 3.3), Python, Pandas, Matplotlib* | [Live](#) | [GitHub](#) 2026
- Built a full-stack AI-powered analytics platform enabling natural-language queries over uploaded CSV datasets
 - Designed a FastAPI backend using Groq LLM APIs to generate insights, tables, and visualization instructions
 - Implemented automated chart generation and deployed the application using React, Render, and modern REST APIs
- RAG + Persona Engine** | *Python, Ollama, ChromaDB, Sentence-Transformers, Streamlit* | [Live](#) | [GitHub](#) 2026
- Built a fully local RAG pipeline that segments long conversation logs using embedding-drift topic detection and fixed-interval checkpointing, summarizing each via a local LLM
 - Designed dual-collection retrieval in ChromaDB combining topic summaries with raw message chunks for grounded, context-aware Q&A
 - Engineered an evidence-grounded persona extraction module and deployed an interactive Streamlit chatbot, running entirely on local inference with no paid external APIs
- Disease Risk Predictor** | *Python, Scikit-learn, Streamlit, Docker* | [Live](#) | [GitHub](#) 2026
- Built an end-to-end ML web app predicting Diabetes and Heart Disease risk from real medical datasets
 - Applied SMOTE for class imbalance and GridSearchCV tuning; achieved 92.44% accuracy on heart disease
 - Containerized with Docker and deployed live on Render
- Network Intrusion Detection System** | *Python, Scikit-learn, TensorFlow, Pandas, NumPy* | [GitHub](#) 2026
- Developed an anomaly-based intrusion detection system using the KDD Cup 1999 cybersecurity dataset
 - Implemented and compared Isolation Forest and Autoencoder models for unsupervised attack detection
 - Achieved **95.81% accuracy**, 98.37% recall, and 91.87% F1-score, significantly outperforming Isolation Forest
- Game Development – IIT Guwahati Game Jam (GameDeon)** | *Godot, GDScript* | [itc.io](#) 2025
- Secured **7th position among 50 teams** nationwide; invited to IIT Guwahati for final showcase

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C++, JavaScript

Data Science & ML: Pandas, NumPy, Scikit-learn, TensorFlow, EDA, Feature Engineering, Model Evaluation, SMOTE, Hyperparameter Tuning

AI / LLM Engineering: RAG Pipelines, Ollama, ChromaDB, Sentence-Transformers, Embeddings, Prompt Engineering, Groq API

Algorithms: Random Forest, Gradient Boosting, Logistic Regression, SVM, Isolation Forest, Autoencoders, TF-IDF, Cosine Similarity

Visualization: Matplotlib, Seaborn

Web & APIs: React, FastAPI, REST APIs, HTML, CSS

Databases & Tools: SQL, MariaDB, SQLite, Git, GitHub, Docker, Linux, Jupyter Notebook, Streamlit

CERTIFICATIONS

IBM Data Science Professional Certificate

Coursera

Tools for Data Science, Data Science Methodology, Python for Data Science, AI & Machine Learning (In Progress)

Google Data Analytics Certificate

Coursera

Data cleaning, analysis, and visualization using spreadsheets, SQL, R, and Tableau

ADDITIONAL

Competitive Programming

Regularly practice DSA on LeetCode and CodeChef (Python, C++); focus on arrays, linked lists, graphs, and DP